

LIVE

Learning Space

NEWSPAPER

22nd MAR, 2019

ITEM WISE EDITORIALS

12th WEEK



RIGHT BALANCE IN HUMAN-TECH INTERFACE !

UPSC CSE – GS III

GS III ... Science and Technology



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NEWSPAPER EDITORIALS

... DISCUSSION

22nd MARCH, 2019

3

Too Much Tech

Boeing 737 crashes suggest erasure of human input in favour of AI is a bad idea

The recent Ethiopian airline crash that claimed the lives of all 157 passengers on board is suspected to have been due to a software glitch on the Boeing 737 Max 8 aircraft. The latter is a new model from the Boeing stable that uses an automated flight software called the Manoeuvring Characteristics Augmentation System. This automatically lowers the nose of the plane when it receives information from sensors that the aircraft is flying too slowly or steeply. However, in the Lion Air crash in Indonesia last October where 189 people on board the same model of aircraft died, the automated software erroneously pulled the plane's nose down at least 20 times as the pilots struggled to manually override the system. A similar issue is believed to have plagued the Ethiopian carrier as well.



There's no denying that advancing technology has made aviation the safest mode of transportation in the world. Yet, as recent air crashes show, technology is a double-edged sword. As the aviation industry moves increasingly towards automated systems driven by a desire to eliminate human error, it seems to account less for technology error. This raises a fundamental debate – are we handing over too much power to technology?

With the advent of artificial intelligence, society is already in the throes of great change. Innovations ranging from self-driven cars to Internet of Things are based on the belief that machines are more reliable than humans. But given that machines are input-driven, wrong inputs produce wrong results. Recall that in 1980 the US and USSR almost engaged in all-out nuclear war due to a false warning produced by a defective chip in American military computers. Greater catastrophes could befall us today if we allow AI to make all decisions without manual override. AI may be today's technological reality. But we must find the right balance in the human-AI interface.



THE TIMES OF INDIA

22nd March, 2019

UPSC CSE – GS III
GS III ... Science and Technology

Awareness in the fields of IT, Space, Computers, robotics, nano-technology, bio-technology and issues relating to intellectual property rights.

IMPORTANT QUESTION
ESSAY/ MAINS / GD / INTERVIEWS

“There is a need to find right balance in the human-artificial intelligence interface”. Do you agree? Discuss with recent examples.

3 Too Much Tech

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RECENT NEWS REPORT

The Crash of an Ethiopian Airlines plane, five months after the Lion Air crash in Indonesia, has forced Boeing to ground all its 737 MAX aircraft.



Two air crashes of Boeing 737 MAX 8 put it in the spotlight.

SALIENT POINTS

- The Boeing 737 MAX series is a single aisle aircraft fitted with high bypass twin turbo fan engines.
- It is locked in competition with Airbus A320.
- For Boeing, this is the model, which is the fastest selling aircraft in its history.
- It cites about 5,000 orders from over 100 customers.
- There are four commercial variants MAX 7, 8, 9, 10.
- The suffix MAX is the coinage meant to highlight the potential of the aircraft in offering “Maximum Competitive Advantage” to the customers.

WHY THE FAILURE OCCURRED?



1

The failure of the aircraft is due to a software glitch on the Boeing 737 Max 8 aircraft.

2

It is the new model from Boeing that uses an automated flight software called the Manoeuvring Characteristics Augmentation System (MCAS).

WHAT WENT WRONG WITH MCAS?



- 1 The software MCAS automatically lowers the nose of the plane, when it receives information from sensors that the aircraft is flying too slowly or steeply.
- 2 In the Lion Air crash in Indonesia, the automated software erroneously pulled the plane's nose down at least 20 times as the pilots struggled to manually override the system.
- 3 A similar issue is believed to have plagued the Ethiopian carrier as well.

TECHNOLOGY IS A DOUBLE-EDGED SWORD!



1 We are increasingly moving towards automated systems driven by a desire to eliminate human error.

2 Are we handing over too much power to technology?

3 Innovations right from self-driven cars, right up to Internet of Things are based on the belief that machines are more reliable than humans.

4 But machines are input driven, wrong inputs produce wrong results.

5 A case in point is, in 1980 US and USSR almost engaged in all out nuclear war due to a false warning produced by a defective chip in American military computers.

Artificial intelligence may be today's technological reality, but we must find the right balance in the human-artificial intelligence interface.

WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao
K. Srikanth
D. Sunil Kumar
L. V. Krishna
Y. Deepthi

Amrita Naidu
M. Binukrishna
G. Ravi Babu
D. Joji
K. Victor Babu

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